



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Constituent of Symbiosis International (Deemed University), Pune

Name of Student: Sumit Kumar Mahato

PRN: 25070122195

Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 11

TITLE: Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Exercise

1. Create an interface **Printable** and implement it in Student and Employee classes.

Users > sumit > Documents > java > java1 > J PrintableDemo.java > Printable

```
1  interface Printable
2  {
3      void print();
4  }
5
6  // Student class
7  class Student implements Printable
8  {
9      public void print()
10     {
11         System.out.println("Student details are displayed.");
12     }
13 }
14
15 // Employee class
16 class Employee implements Printable
17 {
18     public void print()
19     {
20         System.out.println("Employee details are displayed.");
21     }
22 }
23
24 public class PrintableDemo
25 {
26     Run | Debug
27     public static void main(String[] args)
28     {
29         Student s = new Student();
30         s.print();
31
32         Employee e = new Employee();
33         e.print();
34     }
35 }
```

Output

```
[Running] cd "/Users/sumit/Documents/java/java1/" && javac PrintableDemo.java && java PrintableDemo
Student details are displayed.
Employee details are displayed.
```

```
[Done] exited with code=0 in 0.294 seconds
```

2. Create an interface **Switchable** with a method `turnOn()`. Implement the interface in **Light** and **Fan** classes to display the device status.

Users > sumit > Documents > java > java1 > SwitchableDemo.java > Switchable

```
1  interface Switchable
2  {
3      void turnOn();
4  }
5
6  // Light class
7  class Light implements Switchable
8  {
9      public void turnOn()
10     {
11         System.out.println("Light is turned ON.");
12     }
13 }
14
15 // Fan class
16 class Fan implements Switchable
17 {
18     public void turnOn()
19     {
20         System.out.println("Fan is turned ON.");
21     }
22 }
23
24 public class SwitchableDemo
25 {
26     Run | Debug
27     public static void main(String[] args)
28     {
29         Light l = new Light();
30         l.turnOn();
31
32         Fan f = new Fan();
33         f.turnOn();
34     }
35 }
```

Output:

```
[Running] cd "/Users/sumit/Documents/java/java1/" && javac SwitchableDemo.java && java SwitchableDemo  
Light is turned ON.  
Fan is turned ON.  
  
[Done] exited with code=0 in 0.373 seconds
```

Conclusion:

The program successfully demonstrated the implementation of interfaces in Java. The interfaces Printable and Switchable were implemented by different classes, and each class provided its own implementation of the interface methods. This shows how interfaces provide abstraction, flexibility, and a common structure for different classes in Java.